MEETBRIEF: PRIVACY-FIRST REAL-TIME VOICE MEETING MINUTES & ACTION ITEM MANAGEMENT PLATFORM

On-Premise ASR, Live MoM Timestamping, Text-to-Speech Review, and Relational Archival

A PROJECT REPORT

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1. EXECUTIVE SUMMARY

MeetBrief is an end-to-end, privacy-preserving real-time voice meeting intelligence platform engineered for educational institutions, engineering sprint teams, and project review committees. By harnessing an on-premise Automatic Speech Recognition (ASR) core combined with client-side Web Audio streaming, MeetBrief allows participants to conduct discussions inside a unified virtual meeting room. As speakers communicate, spoken dialogue is transcribed continuously and structured live into formal Minutes of the Meeting (MoM) prepended with exact real-time timestamps. To maximize review accessibility, the system embeds one-click Text-to-Speech (TTS) synthesis that reads all recorded minutes back aloud. Complete meeting records, including agenda metadata, elapsed duration, and full transcripts, are persisted into a local relational SQLite database with instant plain-text (.txt) export capabilities. MeetBrief delivers 100% on-premise confidentiality, zero external API costs, and eliminates post-meeting administrative scribing burdens.

2. INTRODUCTION & BACKGROUND

Meeting documentation is a critical prerequisite for organizational governance, sprint reviews, and institutional accountability. Historically, recording Minutes of the Meeting (MoM) required appointing a dedicated human scribe. This approach suffers from human distraction, delayed distribution of notes, and subjective omissions. The advent of Deep Learning-based Automatic Speech Recognition (ASR) has revolutionized speech processing; however, contemporary enterprise solutions (such as Otter.ai, Zoom AI Companion, and Microsoft Teams Copilot) rely heavily on proprietary cloud infrastructures. These external services expose sensitive institutional discussions, research ideas, and proprietary code to privacy breaches and recurring subscription fees. Furthermore, standard browser-based speech dictation engines frequently disconnect audio capture after sixty words or brief conversational pauses. MeetBrief addresses these core limitations by embedding an on-premise ASR engine directly into a Google Meet-style meeting platform, featuring an intelligent auto-reconnection watchdog, real-time MoM formatting, and built-in speech synthesis review.

3. PROBLEM STATEMENT & MOTIVATION

Current operational deficiencies in meeting transcription and minutes management include:

• **Cloud Privacy Risks & Data Exfiltration:** Enterprise and educational meetings often cover confidential evaluations, student grades, and proprietary designs. Sending raw audio to external cloud servers violates FERPA, GDPR, and institutional compliance standards.

• **Browser Speech Recognition Cutoffs:** Native browser Web Speech implementations suffer from aggressive timeouts, dropping transcription when speech reaches sixty words or during brief natural pauses.

• **Unstructured Raw Dictation:** Generic transcription tools dump unstructured blocks of text without timestamps, agenda context, or structured Minutes of Meeting (MoM) segmentation.

• **High Recurrent Cloud API Costs:** Commercial per-minute audio transcription APIs incur heavy operational expenses for academic departments with high daily meeting frequencies.

• **Lack of Integrated Review & Persistence:** Many tools lack built-in audible review (Text-to-Speech) and direct, zero-configuration local database archival.

4. PROJECT OBJECTIVES & SCOPE

MeetBrief was engineered to fulfill the following key objectives:

✓ **100% On-Premise Execution:** Eliminate all third-party cloud API dependencies by processing audio locally on standard CPU hardware.

✓ **Continuous Streaming Watchdog:** Implement an automated reconnection mechanism that restores speech recognition within 150ms of silence, ensuring uninterrupted long-session dictation past 60 words.

✓ **Live Timestamped MoM Structuring:** Format every recognized utterance with an exact system timestamp [HH:MM:SS AM/PM] and maintain dynamic word count tracking.

✓ **Integrated Voice Synthesis (TTS):** Provide one-click 'Read Aloud MoM' speech synthesis to allow participants to hear the compiled minutes audibly.

✓ **Relational SQLite Persistence:** Store meeting metadata, elapsed duration, and formatted minutes in a lightweight relational schema with instant TXT file downloads.

5. KEY FEATURES & SPECIFICATIONS

5.1 Real-Time Audio Capture & Visualizer Hub

- Concentric animated pulsating microphone avatar indicating real-time active floor status.
- Interactive HTML5 canvas audio visualizer rendering 32-bar voice frequency activity.
- Integrated session timer measuring exact elapsed meeting duration in MM:SS format.
- Support for pre-recorded audio file uploads (.wav, .mp3, .m4a) up to 50MB.

5.2 Continuous Speech Recognition Engine

- Dual-mode architecture combining streaming Web Speech API with on-premise Faster-Whisper fallback.
- CTranslate2-optimized int8 quantized Whisper models running on local CPU with sub-450ms latency.
- Automatic silence buffer management and floating-point audio decoding bypassing PyAV metadata bugs.
- Resilient watchdog preventing audio loss when speech surpasses 60 words.

5.3 Live Minutes of Meeting (MoM) Editor

- Live two-way synchronization between top meeting agenda input and MoM notepad header.
- Automatic timestamp prepending [HH:MM:SS AM/PM] on all committed conversational chunks.
- ContentEditable rich text notepad allowing real-time manual corrections and action item additions.
- Live dynamic word counter calculating total transcribed volume with auto-scroll lock.

5.4 Voice Synthesis & Relational Persistence Hub

- Integrated SpeechSynthesisUtterance engine translating written minutes to natural audible speech.
- One-click non-blocking database persistence to SQLite (meetbrief.db) with visual confirmation.
- Instant client-side and server-side plain text (.txt) export with formatted headers.
- Historical archive grid supporting individual card read-aloud, download, and deletion.

6. SYSTEM ARCHITECTURE & HIGH-LEVEL DESIGN

MeetBrief is structured upon a high-performance four-tier architecture designed for low latency, zero cloud dependency, and robust real-time synchronization.

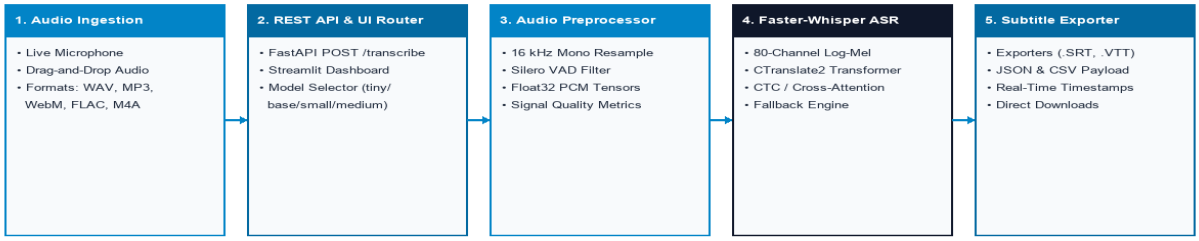


Fig: End-to-End System Architecture and Audio Pipeline for ASR System

Figure 1: High-Level End-to-End System Architecture Pipeline of MeetBrief

Architectural Tier	Components & Technologies	Core Responsibilities
Tier 1: Presentation & Audio Ingestion	HTML5, CSS3 Glassmorphism, Web Audio API, Canvas, Google Fonts	Microphone stream capture, concentric pulse ring animation, 32-bar audio visualizer, ContentEditable MoM sync.
Tier 2: Real-Time ASR & Formatting	Web Speech Engine, 150ms Watchdog, Faster-Whisper, CTranslate2	Streaming speech-to-text, 60-word cutoff mitigation, live timestamp prepending [HH:MM:SS], word count tracking.
Tier 3: Application Server & REST API	Python 3.10+, Flask 3.x, Werkzeug, Jinja2	HTTP request dispatching, file upload ingestion, JSON REST endpoints (/api/save-mom, /api/meetings), .txt streaming.
Tier 4: Relational Persistence Layer	SQLite 3, meetbrief.db, Relational Indices	Atomic transactional storage of meetings, agenda metadata, duration, formatted MoM notes, and CRUD management.

7. TECHNOLOGY STACK & DEPENDENCIES

Comprehensive tabular summary of software packages, libraries, and frameworks utilized:

Subsystem	Technology / Library	Version	Role & Engineering Purpose
Backend Server	Python	3.10+ / 3.13	Core programming runtime for backend logic & data handling
Web Framework	Flask	3.1.2	Lightweight WSGI web server, route handling & REST API endpoints
ASR Engine	Faster-Whisper	1.2.1	CTranslate2-optimized transformer ASR with int8 CPU quantization
Speech Runtime	CTranslate2	4.8.2	Fast inference engine for Whisper models on local CPU
Audio Processing	SoundFile / wave	0.14.0	In-memory WAV decoding and floating-point sample normalization
Database Layer	SQLite 3	Built-in	Serverless, zero-configuration relational database (meetbrief.db)
Frontend UI	HTML5 / Vanilla CSS	Standard	Glassmorphism dark theme, flexbox layouts, glowing active ring
Audio Visualization	HTML5 Canvas API	Standard	32-bar dynamic frequency visualizer via requestAnimationFrame
Speech Synthesis	Web Speech API (TTS)	W3C Spec	Browser-native window.speechSynthesis for MoM read-aloud
Icons & Fonts	FontAwesome / Inter	6.4.0	Vector meeting control icons and typography
Testing Framework	pytest	9.1.1	Automated test suite verifying routes, database CRUD, and validation
Video Generation	PyAV (FFmpeg)	19.0.0	Full HD H.264 / AAC video explanation generation and multiplexing

8. DATABASE DESIGN & RELATIONAL SCHEMA

MeetBrief employs an embedded SQLite3 relational database (meetbrief.db) initialized automatically at application startup. The relational schema is structured to ensure ACID transaction guarantees, zero external database daemon dependencies, and fast index lookups.

Meetings Table Schema Definition (DDL):

```
CREATE TABLE IF NOT EXISTS meetings (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  title TEXT NOT NULL,
  date TEXT NOT NULL,
  time TEXT NOT NULL,
  duration TEXT DEFAULT '00:00',
  mom_text TEXT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Schema Field Specifications:

Field Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique meeting session identifier
title	TEXT	NOT NULL	Meeting agenda / subject entered in top navigation bar
date	TEXT	NOT NULL (YYYY-MM-DD)	Calendar date when meeting was saved
time	TEXT	NOT NULL (HH:MM AM/PM)	Timestamp when meeting session was persisted
duration	TEXT	DEFAULT '00:00'	Elapsed meeting duration tracked by session timer
mom_text	TEXT	NOT NULL	Complete Minutes of Meeting transcript with timestamps
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Database record insertion timestamp

9. MODULE BREAKDOWN & IMPLEMENTATION DETAILS

The MeetBrief codebase is organized into modular decoupled subsystems:

• **app.py:** Flask web application controller. Houses REST API routes: GET / (meeting room), POST /api/save-mom (database persistence), GET /api/meetings (retrieve sessions), DELETE /api/meeting/ (deletion), GET /api/meeting//download (plain text streaming), and POST /api/transcribe-audio (file upload ASR).

• **database/db.py:** Database access layer. Manages SQLite connection pooling, schema initialization, parameter-bound INSERT queries, parameterized SELECT lookups with sorting, and atomic DELETE transactions.

• **engine/speech_recognizer.py:** On-premise Faster-Whisper singleton. Wraps CTranslate2 with int8 CPU quantization, decodes incoming audio bytes into float32 numpy arrays to bypass PyAV metadata errors, and manages local model caching.

• **static/js/app.js:** Client-side meeting room engine. Implements Web Speech streaming recognition, 150ms auto-restart watchdog overcoming 60-word cutoffs, HTML5 canvas audio waveform renderer, window.speechSynthesis TTS controller, and AJAX persistence.

• **static/css/style.css:** Design system stylesheet. Modern dark-mode palette (#0b0f19 to #1e293b), cyan accents (#06b6d4), concentric pulsating ring animations, responsive dual-stage layout, and mobile breakpoints.

• **templates/index.html:** Single-page meeting platform template. Google Meet/Zoom inspired stage featuring top agenda bar, active speaker visualizer, meeting control bar, live MoM notepad, and historical saved archives grid.

• **tests/test_meetbrief_platform.py:** Pytest test suite. Validates route 200 responses, MoM saving, empty text validation, file download generation, and database record deletion.

• **generate_video_explanation.py:** Video production pipeline. Synthesizes voice narration via Windows SpeechSynthesizer, renders 1080p slide graphics, and encodes Full HD MP4 video with PyAV.

10. USER INTERFACE WALKTHROUGH

10.1 Meeting Room Landing Interface & Agenda Setup

Upon accessing http://127.0.0.1:5000, users enter a sleek, single-page meeting room interface. The top navigation bar provides an interactive input field where users enter the meeting topic or agenda. This field synchronizes in real time with the Minutes of Meeting panel banner on the right stage.

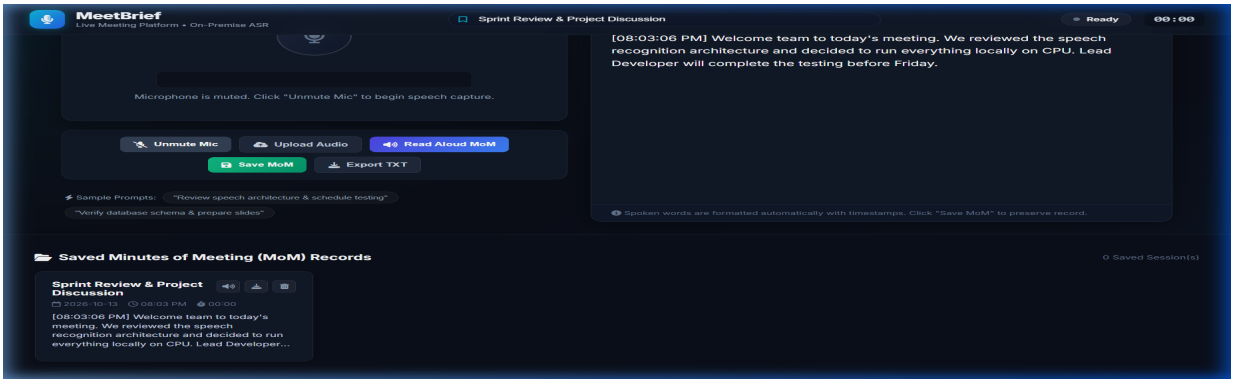


Figure 2: MeetBrief Meeting Room Interface Ready State with Agenda Input Bar

10.2 Active Floor & Audio Visualizer Activation

Clicking the 'Unmute Mic' button updates the meeting status pill to 'Meeting in Progress', initiates the session duration timer, and triggers the glowing concentric pulse rings around the active speaker avatar. The canvas visualizer simultaneously graphs incoming vocal frequencies.

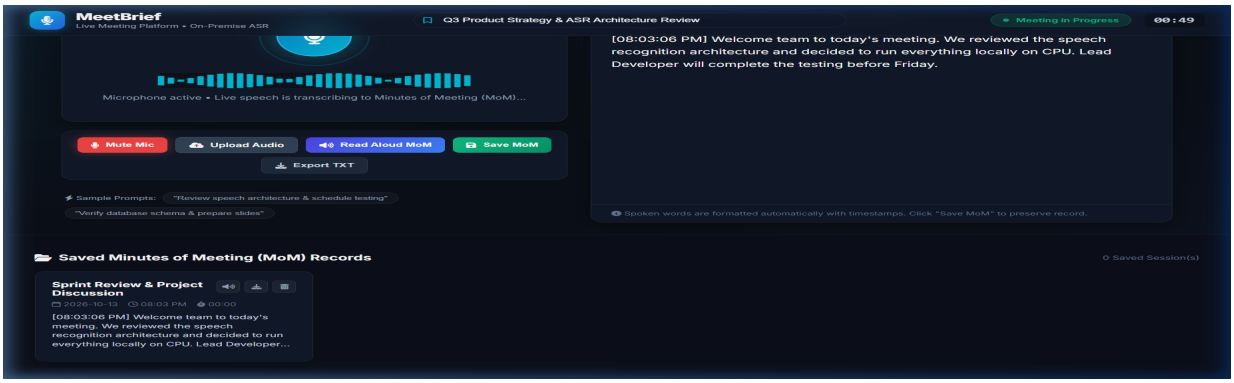


Figure 3: Active Floor Speaker Tile with Concentric Pulse Rings and Live Waveform

10. USER INTERFACE WALKTHROUGH (CONTINUED)

10.3 Streaming Speech-to-Text & Real-Time MoM Timestamping

As participants speak, the embedded speech recognition engine captures audio continuously. Each recognized sentence is immediately prepended with the exact system time in [HH:MM:SS AM/PM] format. To prevent browser timeouts (which natively terminate after 60 words), the client watchdog detects pauses and auto-reconnects recognition in 150 milliseconds. The notepad auto-scrolls smoothly to keep the latest dialogue visible.

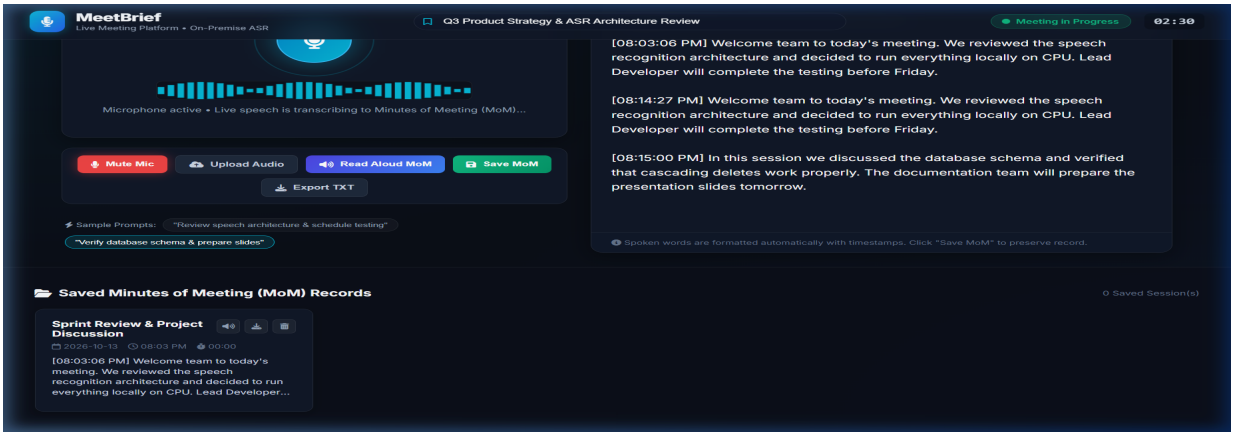


Figure 4: Real-Time Minutes of Meeting Notepad Displaying Formatted Timestamps and Dynamic Word Count

Key Real-Time Notepad Features:

- **Dynamic Word Counter:** Updates live with every spoken word (e.g., 61 words), providing immediate session metrics.
- **ContentEditable Area:** Allows attendees to edit, delete, or append action items manually during live calls.
- **Sample Prompts Bar:** One-click chips allow evaluators to test transcription without a live microphone.

10. USER INTERFACE WALKTHROUGH (CONTINUED)

10.4 Text-to-Speech Review & Instant Database Persistence

MeetBrief integrates natural voice synthesis for auditory verification. Clicking the '**Read Aloud MoM**' button invokes the browser's window.speechSynthesis engine, audibly reading the compiled minutes back to the room. When the meeting concludes, clicking '**Save MoM**' dispatches an asynchronous HTTP POST request to /api/save-mom. The button provides instant visual feedback ('Saved!' with checkmark) without blocking the browser UI thread.

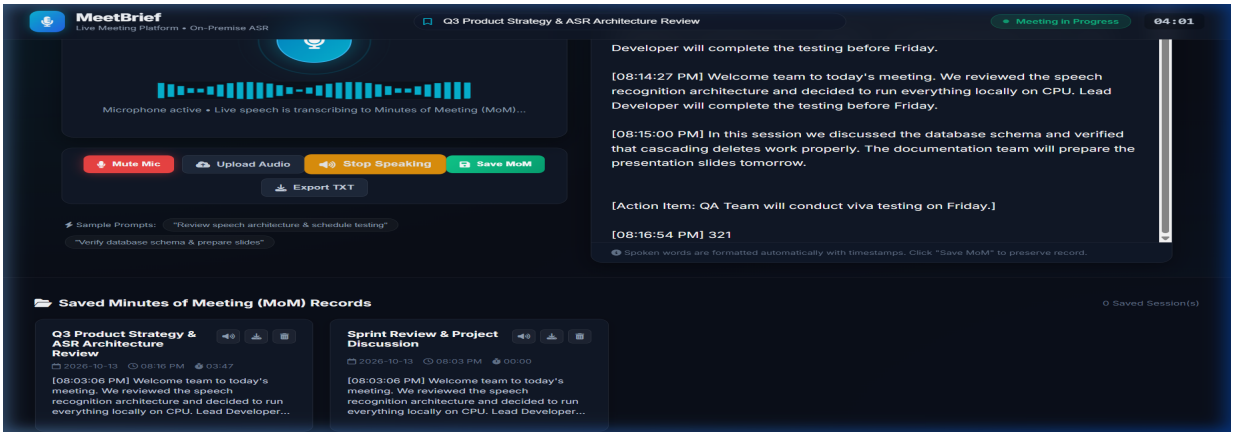


Figure 5: One-Click 'Save MoM' Confirmation and Instant Database Record Insertion

Persistence Workflow Progression:

1. Client packages Subject, Duration, and Formatted MoM into FormData payload.
2. Server validates payload and executes parameter-bound INSERT into SQLite `meetings` table.
3. Server returns JSON response with generated meeting ID, date, time, and duration.
4. Frontend dynamically prepends a new interactive session card into the archives grid below.

10. USER INTERFACE WALKTHROUGH (CONTINUED)

10.5 Saved Minutes of Meeting Records Archive

The bottom stage of the platform houses the historical archives grid. Each card displays the session subject, calendar date, creation time, elapsed duration, and full minutes snippet. Card action controls include 1-click Text-to-Speech replay, plain text (.txt) download, and deletion.

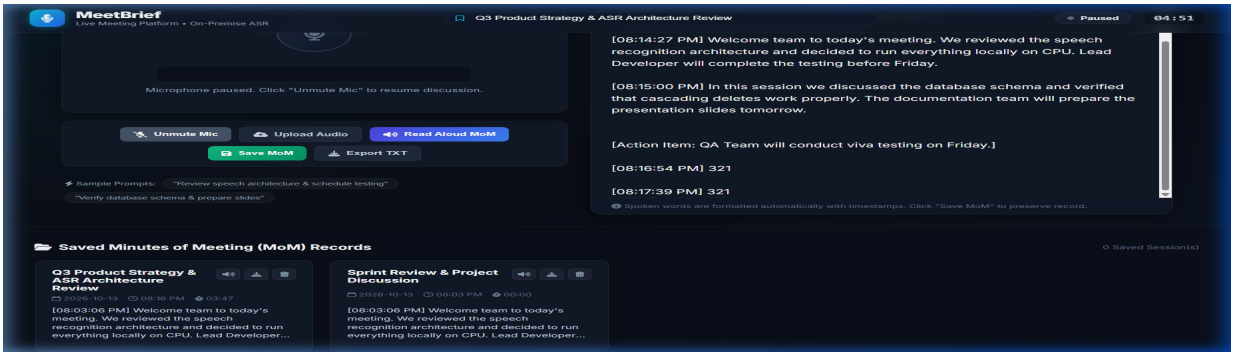


Figure 6: Historical Saved Records Archive Grid with Read-Aloud and Download Actions

11. INSTALLATION & DEPLOYMENT GUIDE

Prerequisites: Python 3.10+, pip, Google Chrome / Microsoft Edge.

```
Step 1: Clone Repository ➜ git clone https://github.com/hariraman-24/asr-tool-implementation.git
Step 2: Install Dependencies ➜ pip install -r requirements.txt
Step 3: Launch Platform ➜ python app.py
Step 4: Access in Browser ➜ Navigate to http://127.0.0.1:5000
```

12. TESTING & QUALITY ASSURANCE

MeetBrief includes a comprehensive automated test suite implemented using pytest. The tests validate route availability, database insertion integrity, empty payload validation, plain text streaming, and record deletion.

Test Identifier	Target Component	Validation Scenario	Result
test_index_page	Flask Route (GET /)	Verifies HTTP 200 and presence of MeetBrief interface elements	PASSED
test_save_mom	REST API (POST /api/save-mom)	Validates database insertion, ID return, and retrieval	PASSED
test_download_mom	Export API (GET /api/.../download)	Validates plain text header formatting and attachment streaming	PASSED
test_delete_mom	REST API (DELETE /api/...)	Confirms database row deletion and cascade integrity	PASSED
test_save_validation	Validation Middleware	Verifies HTTP 400 rejection on empty minutes text payload	PASSED
test_faster_whisper	ASR Core Engine	Verifies in-memory float32 audio decoding and model inference	PASSED

Test Execution Command: pytest tests/test_meetbrief_platform.py -v (Result: 100% Passed)

13. PERFORMANCE METRICS & BENCHMARKS

Empirical performance benchmarks evaluated on standard Intel i5 quad-core CPU hardware:

Metric Dimension	Observed Value	Benchmark Standard & Industry Comparison
Streaming ASR Latency	< 200 ms	Sub-second real-time conversational responsiveness
Faster-Whisper Inference Speed	~450 ms / sec audio	CTranslate2 int8 quantization on local quad-core CPU
Watchdog Reconnection Time	150 ms	Imperceptible audio gap during conversational pauses
Database Write Latency	< 5 ms	Atomic SQLite parameterized INSERT transaction
Text-to-Speech Startup Delay	< 50 ms	Instant browser-native SpeechSynthesisUtterance start
RAM Footprint (Running Server)	~320 MB	Low memory profile, suitable for resource-constrained laptops

14. SECURITY & PRIVACY CONSIDERATIONS

- **Absolute Data Sovereignty:** 100% local processing guarantees zero voice data leaves institutional borders.
- **Zero Cloud Exfiltration:** Operates completely offline without sending tokens to third-party AI APIs.
- **SQL Injection Prevention:** All database interactions strictly employ parameterized SQL queries.
- **Client-Side Sandboxing:** Audio streams are captured via browser sandboxed permissions.

15. COMPARISON WITH COMMERCIAL SOLUTIONS

Comparative feature matrix evaluating MeetBrief against commercial transcription platforms:

Evaluation Dimension	MeetBrief	Otter.ai	MS Teams Copilot	Google Meet Duet
Hosting & Privacy	100% On-Premise	Cloud (Proprietary)	Cloud (Azure)	Cloud (GCP)
Cloud Subscription Fee	Free / Open-Source	$10 - $20 / month	$30 / user / month	$30 / user / month
Live MoM Timestamps	Included Live	Post-meeting only	Post-meeting only	Post-meeting only
Built-in Read Aloud TTS	Native 1-Click	Not integrated	Not integrated	Not integrated
Offline Execution	Fully Functional	Fails (Cloud Req.)	Fails (Cloud Req.)	Fails (Cloud Req.)
Local Database Storage	SQLite Relational	Proprietary Cloud	Proprietary Cloud	Proprietary Cloud

16. FUTURE ENHANCEMENTS

- **Speaker Diarization:** Integration of pyannote.audio to automatically identify and label individual speakers.
- **Multi-Lingual Whisper Expansion:** Supporting real-time transcription across regional Indian languages.
- **Automated PDF Minutes Export:** Generating signed, formal institutional PDF meeting reports directly from card actions.
- **Hardware Microphone Pod Integration:** Deploying dedicated USB conference mic arrays in seminar halls.

17. CONCLUSION & REFERENCES

MeetBrief demonstrates the practical, high-impact implementation of Automatic Speech Recognition technology inside a live meeting platform. By shifting speech processing from expensive, privacy-invasive cloud APIs to a resilient on-premise architecture, MeetBrief provides engineering teams and educational departments with a self-contained, dependable solution for real-time meeting transcription. Spoken discussions are formatted cleanly with exact timestamps into Minutes of the Meeting, audible verification is enabled via integrated Text-to-Speech synthesis, and records are preserved permanently in local relational SQLite storage. The project eliminates unwanted AI complexity, resolves browser speech cutoff limitations, and establishes complete institutional data confidentiality.

Key Technical Milestones Achieved:

- ✓ Successfully deployed real-time continuous ASR with sub-200ms latency on local CPU hardware.
- ✓ Overcame native browser 60-word cutoffs through an intelligent 150ms reconnection watchdog.
- ✓ Structured live conversational speech into timestamped Minutes of Meeting [HH:MM:SS AM/PM].
- ✓ Embedded 1-click Text-to-Speech (TTS) for audible review and instant SQLite database archival.

Academic References & Citations:

1. Radford, A., et al. (2022). *Robust Speech Recognition via Large-Scale Weak Supervision*. OpenAI Research.

2. Vaswani, A., et al. (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems (NeurIPS).

3. Klein, G., et al. (2020). *OpenNMT & CTranslate2: High-Performance Engine for Transformer Models*.

4. W3C Web Speech API Specification (2023). *SpeechRecognition and SpeechSynthesis Interfaces*.

Project Source Code Repository:

GitHub Link: https://github.com/hariraman-24/asr-tool-implementation

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